

Supplementary Information

Notation matters: cross-representation inconsistency in chemistry
language models and its mechanistic origins

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This document provides the verification material referenced in the response to referees: the corrected structural classification and the script that generates it, the model validity screening results, the full per-layer intervention distribution and controls, the robustness analyses, and the outcome of every additional dataset evaluated, including those that failed screening and are therefore not reported in the main text.

S1 Corrected structural classification

The structural-class assignment in the submitted version used `re.search(r'[FClBrI]', s)` for the halogen rule. In regular-expression syntax the square brackets define a character class matching any single character among F, C, l, B, r, I; because C occurs in nearly all organic SMILES, most molecules matched. The corrected rule tests for the element tokens explicitly.

Because assignment is priority-ordered, the defect can only affect classes evaluated *after* the halogen rule. Table S1 gives the full comparison over all 1128 ESOL molecules.

Table S1: Class assignment under the original and corrected classifiers, all 1128 ESOL molecules. The three classes matched upstream of the corrected rule have identical membership.

Class	Original	Corrected	Δ	Position in priority chain
heterocyclic	433	433	0	1st — upstream, unaffected
aromatic	293	293	0	2nd — upstream, unaffected
carboxylic/ester	51	51	0	3rd — upstream, unaffected
halogenated	351	76	-275	4th — the corrected rule
alcohol/phenol	0	179	+179	downstream
aliphatic	0	89	+89	downstream
amine	0	7	+7	downstream
Molecules whose class changes				275 of 1128 (24.4%)

The script below reproduces Table S1 and recomputes all class-stratified statistics from the deposited Phase 1 predictions, without re-running model inference.

```
def classify_original(smiles): # as submitted
    s = smiles
    if re.search(r'[nNsS].*\d|\d.*[nNsS]', s): return "heterocyclic"
    if re.search(r'[a-z]', s) and re.search(r'\d', s): return "aromatic"
    if 'C(=O)O' in s or 'C(O)=O' in s or 'OC(=O)' in s: return "carboxylic/ester"
    if re.search(r'[FClBrI]', s): return "halogenated" # DEFECT
    if re.search(r'(?![a-z])O(?![a-z=\(\)])', s): return "alcohol/phenol"
    if re.search(r'(?![a-z])N(?![a-z=\(\)])', s): return "amine"
    return "aliphatic"

def classify(smiles): # corrected
    s = smiles
    if re.search(r'[nNsS].*\d|\d.*[nNsS]', s): return "heterocyclic"
    if re.search(r'[a-z]', s) and re.search(r'\d', s): return "aromatic"
    if 'C(=O)O' in s or 'C(O)=O' in s or 'OC(=O)' in s: return "carboxylic/ester"
    if re.search(r'[C]l|[Br]|F|I', s): return "halogenated" # FIXED
    if re.search(r'(?![a-z])O(?![a-z=\(\)])', s): return "alcohol/phenol"
    if re.search(r'(?![a-z])N(?![a-z=\(\)])', s): return "amine"
    return "aliphatic"
```

Full script: 09_reclassify_phase1.py in the deposited repository. Run as `python3 09_reclassify_phase1.py phase1_full_results.json`.

S2 Model validity screening

A model that emits a near-constant value irrespective of input will show a sign-flip rate determined by the sign of that constant rather than by any property of the notation, and will appear spuriously notation-robust. Every model is therefore screened on two per-notation diagnostics: the modal fraction (share of predictions taking the single most frequent value) and the correlation between predictions and ground truth. A low count of *distinct* values is not itself evidence of degeneracy, since a model answering on a coarse grid legitimately reuses values.

Table S2: Validity screening. Only the model reported in the main text passes.

Model	Family / precision	Modal frac.	Pearson r	Outcome
qwen3.6:35b-a3b	Qwen3 MoE, native	21.1%	+0.755	pass
nous-hermes2:34b	Llama, 34 B	88.8%	+0.197	fail
qwen2.5:7b	Qwen2.5, 7 B	42.5%	+0.055	fail
hauhau-q4	Qwen3 MoE, 4-bit	—	—	fail (4/40 parseable)
hauhau-iq2	Qwen3 MoE, 2-bit	30.8%	−0.177	fail

Retrospective validation of the reported model. Applied to the deposited Phase 1 predictions, Qwen3.6 gives modal fractions of 19.9% (SMILES), 13.0% (IUPAC), 20.1% (InChI) and 23.7% (SELFIES), with Pearson $r = 0.726, 0.774, 0.753, 0.617$ and Spearman $\rho = 0.776, 0.831, 0.829, 0.662$ (all $p < 10^{-100}$).

Discarded run. nous-hermes2:34b returned a single identical value for all 181 SELFIES prompts and the same value for 88.8% of SMILES prompts. Read naively it yields 98.9% cross-notation inconsistency and a 0% SELFIES sign-flip rate — both artifacts of differing constants per notation. These figures are not reported in the main text.

S3 Full per-layer intervention distribution and controls

Table S3 reports every intermediate layer before any selection, answering the concern that selecting the maximum-improvement layer guarantees a non-negative result.

Table S3: Per-layer intervention effects and controls, all 1051 augmented-SMILES pairs. Improvement in $\log S$; positive values narrow the prediction gap.

Layer	True partner	Random molecule	Reversed	Canonical	% pairs degraded
L_1	−0.006	−0.076	+0.008	0.000000	49.1%
L_2	−0.012	−0.223	−0.086	0.000000	50.0%
L_3	+0.089	−0.458	−0.092	0.000000	44.1%
L_4	+0.038	−0.688	−0.009	0.000000	45.7%
L_5	+0.096	−0.826	+0.000	0.000000	42.9%

Off-manifold check. The canonical control replaces a molecule’s CLS state with its own. It produces a deviation of exactly zero at every layer for all 1051 pairs (0 pairs nonzero), excluding the hook mechanism, CLS substitution as such, and the Ridge head as sources of the observed effects. This does not establish that cross-molecule substitution remains on the activation manifold, and the main text states that limitation.

Specificity. True-partner substitution exceeds random-molecule substitution at every layer, with the margin growing monotonically with depth: +0.070, +0.211, +0.548, +0.726, +0.922 $\log S$ at L_1 – L_5 (Wilcoxon p from 3.2×10^{-10} to 1.1×10^{-80} ; Spearman $\rho = 1.000$ for margin against depth).

Degenerate positions. The embedding position is identical for every molecule (cosine 1.0000, s.d. 0.0000), since the pre-attention CLS state does not depend on the input. The final position is tautological as an intervention: the prediction head reads it, so replacing it substitutes the source molecule’s prediction wholesale, and the resulting “improvement” equals the unpatched gap to within 5×10^{-5} for all 1051 pairs. Both are excluded.

S4 Robustness of the behavioural effect

Table S4: Sensitivity of the inconsistency rate to the spread threshold ($n = 1072$), and to reformulating the task as classification ($n = 790$ complete cases).

Threshold ($\log S$)	% inconsistent	Classification scheme	% disagreeing
0.10	94.5%	4-band	71.8%
0.25	93.6%	3-band	45.9%
0.50	88.0%	2-band (binary, $\log S = -2$)	27.8%
0.75	80.9%	Band accuracy, binary: 82–90% across notations	
1.00	61.0%		
1.50	46.9%		
2.00	26.3%		

Metric. The standard deviation across all four notation-specific predictions has mean $0.830 \log S$ (median 0.700) and ranks molecules almost identically to the range (Spearman $\rho = 0.987$, $p < 10^{-300}$, $n = 790$). Under the standard deviation on the matched set, halogenated compounds are marginally more inconsistent than carboxylic/ester compounds (1.094 against 1.001), so the identification of a single most-inconsistent class is sensitive to metric and subset; the main text states the carboxylic/ester result as holding for the full sample under the range metric.

Non-language-model baseline. A RandomForest on eight RDKit descriptors (5-fold CV MAE $0.237 \log S$) shows a mean prediction spread of $0.105 \log S$ across four string routes to the same molecule, against $1.686 \log S$ for the language model, with 9.8% of molecules exceeding the $0.5 \log S$ threshold against 88.0%. For 79.2% of molecules every string route yields a bit-identical Morgan fingerprint and therefore an identical prediction.

S5 Additional datasets: all outcomes

Table S5: Every additional dataset evaluated, including those that failed the validity screen and are therefore not reported as evidence in the main text.

Dataset	Task	Screen	Outcome
FreeSolv	regression	pass ($r = 0.646, 0.599, 0.460$)	effect replicates; see below
Lipophilicity	regression	fail ($r = 0.180, 0.163, 0.200$)	not reported
BBBP	classification	fail (below majority class)	not reported
BACE	classification	fail (below majority class)	not reported

FreeSolv replication. 81.9% of molecules exceed a 0.5-unit three-notation spread, against 83.6% for the same three notations on ESOL. Mean spread is $3.364 \text{ kcal mol}^{-1}$ and $1.404 \log S$ respectively; normalised by the range of each target this is 14.3% and 13.0%. Notation ranking does not transfer: InChI has the lowest error on ESOL and the highest of the three on FreeSolv (MAE 3.054 against 2.596 for SMILES).

Classification datasets. On BACE the model answered positive for 88.7% of SMILES prompts but 8.0% of SELFIES prompts, at 50.0% accuracy against a 60.0% majority class. Read naively this gives 90.0% cross-notation label disagreement. That figure is not reported, because a model that cannot

perform a task cannot be used to measure notation robustness on it. Whether the effect extends to molecular classification therefore remains open.

All scripts, per-molecule predictions and control outputs are deposited with the article; see the Data availability statement.